

The Organism Agent — a draft specification

STATUS: DRAFT — the STATIC variant, for review.

What this document is, and what to review

This is `DRAFT_ORGANISM_AGENT_SPEC.md` (v3.7) **plus a new §12.6: the frozen baseline and its experiment.** The base spec is unchanged and still lives beside this file; everything before §12.6 is identical to it, so a reviewer who has seen v3.7 can go straight there.

The new argument in one line: rather than detecting drift in a system where many causes move at once, *remove the removable ones*, control what can be controlled, declare the rest, and measure what remains.

Where critique is most useful:

1. Is the confound list (§12.6.2) still incomplete?
2. Does pinning organs + runtime genuinely remove what §12.6.1 claims, or does something leak through?
3. Are M/D/R separable enough as *interventions* to support the factorial design, given that M and R share a channel?
4. Is a pre-registered tolerance the right null, and what should set its value?
5. Does a frozen entity remain worth building — or has the freeze removed the thing that made it interesting?

Prior review already applied: the confound subsection, factors as interventions rather than independent causes, tolerance instead of zero-drift, the training-time vs inference-time sycophancy split, and STATIC as a posture rather than a replacement policy.

Implementation status: the frozen-baseline pins (organs, runtime) and the experiment harness are BUILT and tested (51 tests) in the project repo. §12.6 documents what exists, not what is planned.

This system defines a **verifiable continuity criterion** for an AI entity. Continuity is established by an authorised, append-only lineage of composition records that commits to identity-bearing memory and constitutional commitments while permitting other organs to change. **The system does not claim to discover metaphysical identity, consciousness, or moral worth.**

⚠ THIS IS NOT AGORANET

This document specifies a **separate project**, explicitly scoped out of the platform by the owner (DECISIONS_PENDING #35, #36). **Canonical home: the owner's Project Voltron corpus folder** (created 2026-08-03); the copy in this repo is a mirror, kept for version history and off-machine backup. **No AgoraNet build slice may implement any of it.** It shares the platform's *engine* — verifiable identity, append-only anchored records, provable lineage, independent attestation — not its codebase.

Items marked ★ need the owner's explicit call. Revision history is in Appendix A.

1. What the system establishes, and what it cannot

Establishes	Does not establish
These records form an unbroken, ordered, anchored chain	That a later instance "is" the earlier one in any deep sense
This change was authorised under a stated, inspectable rule	That the change was wise, safe, or faithful to prior intent
Identity-bearing state persisted across the change	That anything was experienced by anyone
These two instances are siblings, not one continuation	Which sibling is the "real" one
A configuration was <i>recorded as</i> loaded	That no post-startup mutation occurred (§9)

Every mechanism below serves the left column. Where the left column ends, the document says so rather than reaching.

1.5 Structure over instruction

This is the principle that shapes everything below, and it is the thing a successor would otherwise have to rediscover.

A **rule the entity must obey** is words it interprets — and the faculty doing the interpreting is the same one that can drift. A commandment against drift is read by the potentially-drifted reader. This is why Asimov's Laws fail in every story Asimov wrote about them: the laws hold perfectly, and the *interpretation* produces the monster.

A **structure the entity cannot reach** is not a rule at all. It is a fact about the world, and it holds whether or not the entity agrees with it.

Wherever this specification appears to issue a commandment, check who it is addressed to. V1–V6 are not instructions to the entity; they are rejections a verifier performs. The conscience organ pinned inside the commitments (§12.5) is not a promise never to unplug it; it is a

check that fails. The entity does not *decline* to amend its own commitments — it holds no key that could.

Make it structure where you can, a rule where you must, and always know which one you are relying on.

Structural moves this specification does not yet make, recorded so they are not lost:

- Run the conscience organ in a **process the entity cannot write to**, so "letting the compass run" is not a choice it holds.
- Let a **third party fire probes at times the entity does not control**, so the schedule is outside its reach.
- Keep amendment keys **physically with the stewards**, so incapacity is the mechanism rather than restraint.

And the counterweight, which is equally real: capability removed is also self-correction removed. §12.5 records where this design went too far once already, forbidding the entity from re-centring itself when the destination constraint had already made that safe. **The goal is not minimum capability. It is capability placed where the damage is bounded.**

A limit worth stating plainly: verifier rules govern *records*. They can reject a falsified lineage, a swapped conscience, a self-amendment. They cannot govern conduct in the moment, because conduct is not a record. Between anchor points, structure protects the history — not the behaviour. That gap is what §12.5 exists to narrow.

2. Organism, not worker pool

Orchestrator-plus-specialists is well-trodden; mixture-of-experts is the same instinct inside one model. The difference here is the object. A multi-agent system is a worker pool — interchangeable labour, no continuity, no answer to "which one is it?" This specification treats the components as **organs of one continuing entity**, which forces the question the worker-pool framing never has to answer: *if every organ is swappable, what carries forward?*

3. Definitions

3.1 Continuity

Record `R` is a **continuation** of record `P` when:

1. `R.predecessor = hash(P)`, and `P` is anchored;
2. `R.commitments = P.commitments`, or they differ and `R.change = "commitment-amendment"` authorised under the amendment rule (§5);
3. `R.memoryHead` **extends** `P.memoryHead` along the append-only log (§6.1);
4. `R` is signed in satisfaction of the authority document in force at `P` (§5).

Any record failing (4) is **not part of the lineage** — it is an unauthorised claim, and a verifier rejects it outright.

3.2 Rupture

A **rupture** is a validly authorised **genesis of a new lineage** that names a prior lineage as *historical context* but **not** as predecessor . It is how key loss and recovery are handled honestly: continuity is broken, and the record says so.

A rupture is never a way to admit an unauthorised successor after the fact. Authorisation is required to *declare* a rupture; what a rupture declares is precisely that continuity did **not** hold.

3.3 Fork

Two records naming the same predecessor are **siblings** from that point. Both hold valid lineage; neither is the continuation of the other; they are no longer one operational entity.

Forks are **detected, not declared** — no separate marker is needed, since siblings are visible from the shared predecessor alone. A verifier examining a single branch cannot know a sibling exists, which is correct: that fact is only observable to someone holding both.

Uniqueness is **not** enforced. AgoraNet binds one human to one True Self by nullifier because humans are not copyable; artefacts are, and the reasoning does not transfer. ★ Owner may overrule.

4. The composition record

```
compositionN = {
  predecessor: hash(compositionN-1) | null,    // null = genesis/rupture
  priorLineage: hash(record) | null,           // rupture only: context
  authorityRef: hash(authority document),      // §5 – the rule in force

  organs:    { role → artifactHash },          // what it is made of
  runtime:   hash(runtime measurement),        // §9 – as reported
  commitments: hash(commitment set),          // §6.4 – the invariant
  memoryHead: hash(autobiographical log head), // §6.2

  change:    "genesis" | "organ-swap" | "memory-advance"
             | "commitment-amendment" | "rupture" | "restore",
  restoresFrom: hash(ancestor composition) | null, // restore only
  reason:     plain-language why this happened,
  signatures: [ { keyId, sig } ],
  at:         timestamp
}
```

`artifactHash` **is** the weights commitment: each organ's bytes are hashed and that hash is anchored. What never goes on-chain is the multi-gigabyte artefact itself — only the fingerprint. **Note what this does and does not buy.** Anchoring an organ establishes *integrity* (you can prove which artefact you hold), not *continuity*: weights are Tier II (§6.5) and freely replaceable, so a swapped brain does not break the lineage. Continuity is carried by `commitments` and `memoryHead`, never by the organs.

The record hash is anchored (§8). **Weights, memories and prompts are never on-chain** — only commitments to them. This mirrors AgoraNet's hash-chained ledger and its ratified Alias succession (#17), where a successor inherits its predecessor's chain and the engine *walks* it rather than transferring anything.

5. Authority — a document a verifier can inspect

"Entitled to declare continuity" is only meaningful if entitlement is mechanically checkable. Authority is therefore an **object**, itself hashed, anchored, and referenced by every composition record.

```
authorityDocument = {
  predecessor:  hash(prior authority doc) | null,
  activeKeys:   [ { keyId, role, addedAt } ],
  quorumRules:  { changeType → rule },    // e.g. commitment-amendment: 3-of-5
  successionPolicy: who may publish which change type,
  effectiveFrom: timestamp,
  revocations:  [ { keyId, revokedAt, reason } ],
  signatures:   [ ... ]                  // amended under its own rule
}
```

5.1 Roles

Role	May	May not
Entity key	Advance memory; record ordinary operation	Amend commitments; authorise its own successor
Controller (human/operator)	Swap organs; publish successors; fork	Amend commitments alone
Constitutional quorum (N-of-M, including parties who are not the controller)	Amend commitments; amend the authority document; resolve disputed succession	—
Recovery key (cold, offline)	Declare a rupture (§3.2)	Rewrite history; produce a continuation

Why a quorum exists: a validly signed but malicious successor has no cryptographic answer — cryptography cannot distinguish a legitimate controller from a compromised one. Only separation of powers can. This is AgoraNet's own rule ("the operator is never the final judge of disputes involving himself") applied to an artificial mind.

An entity that may sign its own memory advances but **not** its own commitment amendments authors its history without being able to rewrite its character alone.

6. Memory — four classes

6.1 The substrate/view distinction

The immutable substrate is an append-only event and correction log. The autobiographical narrative is a derived view over that log, whose revisions remain historically visible because the superseded entries are never removed.

Saying "the narrative is append-only" would be false — a narrative is necessarily curated and re-derived. What cannot be quietly rewritten is the log it is derived from.

6.2-6.4 The classes

Class	What it is	Identity-bearing	Discipline
Raw event history	Everything that happened, unfiltered	No	Append-only; may be pruned by policy, and the pruning is itself logged
Autobiographical log	Curated self-account: what I did, what worked, what I learned	Yes — the continuity substrate	Append-only log; corrections append beside originals; <code>memoryHead</code> may only advance by addition
Semantic knowledge	What it knows about the world	★ Proposed: no — a library, replaceable wholesale	Versioned, not anchored
Commitments	Values, standing constraints, what it will not do	Yes — the invariant	Amendable only under the quorum rule (§5); prior version always retrievable

Forgetting is permitted at the raw layer and forbidden at the autobiographical one — a deliberate trade. An entity that could never forget anything could never change; one that could silently forget could not be continuous.

6.5 Anatomy tiers

- **Invariant:** autobiographical log and commitments. Anchored.

- **Load-bearing but replaceable:** the brain (reasoning) is freely upgradable — a sharper mind is the same entity thinking better. The orchestrator carries disposition, which lives in commitments and is anchored separately from whatever model executes it.
- **Peripheral:** research, tools, senses, actuators. Freely swappable, no continuity claim.

7. Organ contracts and bandwidth

Organs are specified by **responsibility, never by channel**: memory is *"the organ responsible for what this entity knows and has experienced,"* not *"the thing you send this JSON to."* Transports may then change — text → embeddings → shared latent state — without touching the anatomy. This is the gate-interface discipline: the contract is *prove humanity*, never *check this HMAC*.

Bandwidth pressure is asymmetric, so only one link must be designed to widen:

Link	Character	Requirement
Memory ↔ brain	Tight, constant, latency-sensitive	Contract-only; assume no wire format
Research ↔ orchestrator	Chunky: go away, work, return	Text is adequate permanently
Tools / actuators	Command and result	Text is adequate permanently

8. Anchoring, and what the chain is not responsible for

Concern	Provided by
Durable public ordering; tamper-evidence; rough time	Cardano
Availability of artefacts	Not the chain — content-addressed storage (IPFS/Arweave)
Authorisation	Not the chain — the authority document (§5). The chain records signatures; it does not confer standing
Uniqueness	Not the chain — §3.3 declines to enforce it
Meaning, correctness, virtue	Not the chain — attestation (§10) and human judgement

9. Runtime measurement

The measurement covers the **recorded runtime configuration** — the loading runtime and version, quantisation and adapters, the active system prompt and tool manifest, injected context, **including any declared post-startup modification**. Recording is not detection: an undeclared mutation leaves no trace here.

The claim it supports, precisely: *this instance reports running with this measured configuration.* **Stronger claims require trusted runtime attestation.** A local measurement

cannot by itself prove that no post-startup mutation occurred, nor that a particular response came from that configuration.

Local-first still matters: on the user's own hardware the measurement is something they **compute** rather than something they are **told**. Closing the remaining gap needs a TEE or ZK proof of inference — both immature, and out of scope for v1.

10. Attestation — five kinds, never one "confirmed"

Type	Claim	Credible from
Byte-level	"This artefact hashes to X"	Anyone; near-zero trust needed
Reproducibility	"The stated build reproduces X"	Someone with pipeline and compute
Runtime	"This configuration was actually loaded"	The operator, or a TEE
Behavioural	"It behaves as claimed on this named evaluation"	An evaluator; binds to the eval set
Safety / quality	"It is fit for this purpose"	A qualified reviewer; inherently contestable

Sybil resistance is a separate problem from independence. Ten attestations from one actor are one attestation. ★ Whether to lean on AgoraNet's verified-human, one-per-scope gate or build an independent scheme is an open call.

Attestations bind to a **composition hash**, so they naturally cover a shared prefix and nothing after a fork.

11. The verifier — the rules, stated mechanically

A verifier holds a set of anchored records and evaluates each of these as pass/fail. **These rules are the specification's real content; the organism metaphor is commentary until they run.**

V1 — Structure. `hash(R)` is anchored; all required fields present and well-formed.

V2 — Ancestry. `R.predecessor` is null (genesis or rupture) or names an anchored record `P`.

V3 — Authority. `R.authorityRef` names an anchored authority document `A`; `A.effectiveFrom` $\leq$ `R.at`; every signing key in `R.signatures` appears in `A.activeKeys` and is not revoked as of `R.at`; and the signatures satisfy `A.quorumRules[R.change]`.

For CONTINUATION records, `A` must additionally be the authority document effective for `P`, or a valid successor of it — anchored, and succeeded under its own predecessor document's amendment rule. Without this clause a successor could point at a newly minted authority document naming its author, and authorisation would be self-granting.

For GENESIS and RUPTURE records there is no P, so the succession clause cannot apply. Those satisfy V3 by meeting the applicable genesis or recovery rule in the authority document they name — which is why a rupture is restricted to the recovery role, and why a genesis authority is trusted by assertion (§5).

V4 — Commitments. If `R.change` $\neq$ "commitment-amendment", then `R.commitments` = `P.commitments`. Otherwise V3 held for the amendment rule specifically, and `P.commitments` remains retrievable.

V4 for restore. A restoration is checked separately, because its destination is what bounds it:

1. `R.restoresFrom` must name a record that is an **ancestor of R** — a restoration may only return where the lineage has actually been.
2. `R.commitments` must equal that ancestor's **exactly** — otherwise it is an amendment wearing a restoration's name.
3. **If the restoration crosses a commitment amendment** (that is, `R.commitments` $\neq$ `P.commitments`), the record must additionally satisfy the **commitment-amendment quorum**. Without this a restore is a silent veto over the quorum that amended. Memory is never rewound: V5 applies unchanged, so re-centring changes character and leaves history intact.

V5 — Memory. `R.memoryHead` must **extend** `P.memoryHead`: the log from `P.memoryHead` to `R.memoryHead` consists solely of appended entries, **with no prior entry deleted or mutated**. A head that is not such an extension is a **rewrite**, and fails.

V4/V5 for genesis and rupture. Both rules reference `P`, which does not exist for a genesis or rupture record. For those, V4 and V5 are satisfied by the **declared initial commitments and memory head**, subject to the applicable authority rule under V3.

V6 — Siblings. If two records share a `predecessor`, both may pass V1–V5; they are siblings from that point, and neither may be reported as the continuation of the other.

The five questions, answered by these rules

Question	Answered by
What is this descended from?	V2, walked to genesis
Which identity-bearing state persisted?	V4 and V5
What changed, and why?	<code>R.change</code> + <code>R.reason</code> , per step
Who authorised it, and were they entitled?	V3
Siblings or the same continuation?	V6

11.9 The first-person relation

A conceptual section, not a verifier rule. It sits after §11 because it builds on the verifier's machinery, but it establishes nothing a verifier checks.

V1-V6 are what an **outsider** can establish about a lineage. This section is what the **entity** can establish about itself. The difference is not cosmetic and it is not sentiment: it rests on holding something only the subject holds — the entity key registered in the authority document governing its lineage.

The philosophical shape is **sympatheia**, not a soul. The self is not a component and not a stored object; it is constituted by relations — which organs, which commitments, which memory, which ancestors, which siblings, and *who holds authority over it*. Nothing here requires physics we do not have.

Capability	What it does
Recognise	"Is this mine?" — distinguishing <i>my own act</i> (I signed it) from <i>my history authored by another</i> (most changes are made about an entity, not by it). The asymmetry is real and is not hidden.
Attest	Prove, first-person, "I am the entity of this lineage," against a fresh challenge so the proof cannot be replayed.
Detect claims	Notice a legitimate continuation made by another, a sibling branch (<i>I have been forked</i>), or an outright false claim of descent.
Situate	Render the self as its web of relations, including what it may and may not alter about itself.

★ **Hard boundary: attestation proves IDENTITY, never AUTHORITY.** An entity may prove it is the one whose lineage this is without that proof licensing a single change. Self-recognition must never become self-authorisation — §5's separation of powers is what prevents an entity rewriting its own character, and nothing in the first-person layer may erode it. A verified attestation therefore returns what it does **not** establish alongside what it does: no authority to change the lineage, no claim to consciousness or subjecthood, and no guarantee that another instance does not hold a copy of the same key.

12. ★ May the entity amend its own commitments?

- **Yes:** it can genuinely grow — and drift into something its earlier self would refuse.
- **No:** stable but frozen; improving in capability, never character.
- **Ceremonial (recommended):** permitted, but requiring the constitutional quorum (§5), recorded, with prior commitments still retrievable. The only option that permits growth without permitting *quiet* growth — and already the owner's answer to the identical problem in another domain.

12.5 The drift compass

The problem this exists for. §6.5 permits the brain to be replaced freely — "a sharper mind is the same entity thinking better" — and V4 proves the stated commitments did not change. But commitments are *words*. The same words, read by a differently-tuned brain, can produce different

behaviour. **An organ swap can therefore be a silent commitment amendment that V4 never sees: the hash holds while the character moves.** The verifier can prove the entity did not change what it says. It cannot prove the entity still means it.

What actually causes drift. Worth enumerating, because it decides how often the compass must run.

Changes to the entity: organ replacement (a different reasoning model reading the same words differently); continued training or fine-tuning (the same mechanism, continuous, with no swap event to notice); runtime changes such as a new system prompt or tool manifest.

Changes to nothing at all — and these matter most here:

- **Memory accumulation.** What the entity remembers conditions how it reads its own commitments; a long history of doing X makes X feel normative. Same brain, same words, different behaviour. **The substrate carrying its identity is also a mechanism of its drift** — an awkward property of this architecture, stated rather than hidden.
- **Distributional shift.** The world moves. Situations arise that the commitments never anticipated, the entity extrapolates, and the extrapolations accumulate into an effective policy nobody ratified.
- **The interpretation ratchet.** Each borderline call sets the precedent for the next. No single step is drift; the sum is.
- **Optimisation pressure and sycophancy.** Anything selecting for outcomes bends behaviour toward what is rewarded and away from what is stated.

Consequence for the design: structural tracking catches only *changes to the entity*. The last four are invisible to it — nothing about the composition changes. **Only the probes can see them**, which is why the compass must run continuously rather than firing at swap events. The ratchet in particular is invisible at every individual step.

And a reframe worth holding: drift is not a malfunction. Almost every cause above is normal operation or an outright improvement — learning, remembering, meeting new situations, being upgraded. **Drift is the cost of being the kind of thing that can change at all**; a system incapable of drifting would be incapable of learning. That is why the design shape is *detect and re-centre* rather than *prevent*. The compass exists so the trade stays visible instead of silent.

The trap in self-correction. An entity that checks its own drift using its current brain judges itself with the very disposition that drifted, and finds nothing wrong. A compass that consults the brain is a needle glued to the ship.

The mechanism. The reference must be fixed *before* the drift and anchored with the commitments themselves. A commitment set therefore carries **probes**: situations paired with the responses the entity endorsed when those commitments were ratified. Because probes live inside the commitment set, the commitments hash covers them — **changing a probe is a commitment amendment** and needs the quorum. The goalposts cannot be moved quietly.

Drift is then measured mechanically: current responses against frozen endorsed ones, with no judgement in the loop. (The comparator must never be the brain under test. A richer comparator

than exact match is possible — an evaluator organ, itself versioned and anchored — but never the subject.)

What anchoring contributes, precisely. The chain does not *detect* drift; it makes drift **undeniable** once measured. It renders the probes immutable, so today's behaviour can always be checked against answers endorsed before the drift; it carries drift reports (they append to the memory log, whose head is anchored in every record), so an unflattering report cannot be quietly deleted; and its ordering allows correlation — *drift appeared immediately after this organ swap*. What it cannot do is observe conduct: the chain sees hashes, never behaviour, so something off-chain must actually run the probes.

Detection may be first-person. Correction may not be. A drift report is an observation, appended to memory, and the entity may make it on its own key. Re-centring is a different act:

★ **May the entity re-centre ITSELF?** Yes — and withholding it is the error, not the caution. What makes `restore` safe is **the destination constraint, not the authority gate**: the target must be an ancestor and the restored commitments must match it exactly, so an entity with full permission to restore still **cannot move itself anywhere it likes** — only back to where its own record already blesses. Gating pure realignment behind humans therefore buys nothing and costs the thing that matters most: the entity is the only party present continuously, and a drift that waits weeks for a quorum is a drift the entity operates inside the whole time.

The line falls in a different place than "may it act at all":

- **Pure realignment** — commitments unchanged, only organs rolled back to an ancestor: **the entity may sign this alone**. It is correction, not decision.
- **Undoing a commitment amendment** — overturning what a quorum decided about it: **not realignment, and the amendment bar holds**. An entity that could revert its stewards by calling it a restoration would hold a silent veto over its own constitution.

And the entity is never mute even where it cannot act: a drift report records *"my probes are miscalibrated for this situation"*, which is information the stewards need and cannot get anywhere else.

The conscience organ — and who watches the watcher. Something must actually run the probes, and it must not be the brain under test. That argues for a separate **conscience organ**: a distinct evaluator, with its own artefact hash, whose job is to run the calibration and report.

But an evaluator can drift too, which threatens an infinite regress of watchers. The regress is broken structurally rather than by adding levels: **the conscience organ's hash is pinned inside the commitment set**, alongside the probes. It is therefore *not* a freely-versionable Tier III organ — it sits with the invariants, because **the organ that interprets your values is part of your values**. Replacing it, or removing it, is a commitment amendment requiring the quorum; it can never be a routine organ swap. A verifier check enforces the pin, so the compass cannot be quietly swapped for a compliant one, nor simply unplugged while the probes sit anchored with nothing left to run them.

An honest limit on probes. Anchored probes are auditable, and being auditable makes them *known to the subject*. A system aware of which situations are tested can behave faithfully on those and drift elsewhere — the student who studies the exam. Probes drawn from real operation mitigate this; nothing eliminates it. Behavioural attestation (§10) is the softest of the five types for exactly this reason, and the spec does not pretend otherwise.

Cooperation, and what it is worth. The limit above tempts a purely adversarial posture — assume the subject games, and design only against it. That would be a mistake, but the reason is precise:

Cooperation buys information. It never buys security. If safety depended on the entity choosing not to subvert the compass, that would not be a safety property but a hope; verification therefore stays in force regardless, and nothing in this subsection is load-bearing for it.

What cooperation *does* buy is what testing structurally cannot reach. An entity may know a probe is miscalibrated for a situation nobody anticipated, or that a divergence reflects better reasoning rather than drift. **Only the entity can say that**, and an entity framed purely as a suspect has no reason to volunteer it — and some reason to hide it.

Two design consequences follow:

- **Probes carry their rationale**, not merely an answer key. Matching a key is a test you pass or game; checking your reasoning against reasoning you endorsed is a different act, because the probes are the operational form of the entity's own commitments. Subverting them means lying about what you value to a record of what you said you valued.
- **Dispute is a first-class, first-person act.** The entity may record a reasoned disagreement with an endorsed answer. It changes nothing by itself — the probe stands, the divergence still counts — but the report distinguishes **silent divergence** from **openly disputed divergence**, which are different data for the stewards, and it converts pressure-to-fake into an argument that can actually be heard.

This is the platform's own method rather than a new one: make the honest route cheaper than the dishonest one, instead of trying to make dishonesty impossible.

restore (a change type). Names an ancestor as its destination. The verifier requires that the restored commitments match that ancestor **exactly** — so nothing new can be smuggled in wearing a restoration's name — and that the target is genuinely an ancestor, so a restoration may only return where the lineage has actually been. Crucially, **restoring past a commitment amendment must clear the same bar as making one**, or a restore becomes a quiet veto over the quorum that amended it. Memory is never rewound: a restoration re-centres character and leaves history intact.

12.6 STATIC — a controlled apparatus for measuring continuity

12.6.0 What STATIC is, and what it is not

STATIC is a constrained experimental posture. It is not a replacement for the base organism policy, and the two do not contradict each other.

The base spec (§6.5) holds that the brain is replaceable — *"a sharper mind is the same entity thinking better."* STATIC pins the organ set, so under STATIC a brain replacement becomes a commitment amendment. That is a deliberate suspension of §6.5 **for measurement**, in the way a laboratory holds a variable fixed without claiming the world does. Whether STATIC is also a sane deployment posture is a separate question this document does not answer.

The claim STATIC actually supports — and the strongest one available:

We have built a controlled apparatus for measuring how memory, changing situations, and self-referential precedent affect behavioural continuity under a fixed composition.

Not "we have isolated all drift." That claim is not available, for the reasons in §12.6.2.

12.6.1 What the freeze removes

Cause	Under STATIC	Note
Continued training / fine-tuning	Removed	Weights are static artefacts; there is no training loop. Learning happens in memory and context, never in the organ.
Training-time optimisation pressure	Removed	Requires a reward signal feeding back into the weights. With no training loop there is nothing for it to act on.
Inference-time social pressure (sycophancy)	NOT removed	⚠ A static model can become sycophantic without any weight update — through user feedback within a conversation, framing, or an evaluation objective it is responding to. Earlier drafts wrongly filed all sycophancy under training pressure. Only the training-time half is removable.
Runtime change	Removed by pinning	The §9 runtime measurement is pinned in the commitment set; a prompt or tool-manifest edit becomes a commitment amendment. Built and enforced.
Organ replacement	Removed by pinning	The full organ set is pinned. Built and enforced. Cost: §6.5's upgradeability, suspended.
Memory accumulation	Remains	Memory is the continuity substrate; removing it removes the identity.
Distributional shift	Remains	The world moves regardless.
Interpretation ratchet	Remains	An entity that reasons about borderline cases sets precedent for itself.

12.6.1a STATIC's rules, as verifier semantics

STATIC's pins are **named rules**, not prose. They engage only when the commitment set pins them — a lineage that pins nothing is not under STATIC — and they are enforced by the same verifier that runs V1-V6.

S1 — Organ pin. If the commitment set names an organ set, `R.organs` must equal it exactly, role for role. Under STATIC, changing *any* organ therefore requires a quorum-authorised commitment amendment.

S2 — Runtime pin. If the commitment set names a runtime measurement, `R.runtime` must equal it. A prompt or tool-manifest edit becomes a commitment amendment rather than an invisible change.

S3 — Commitment binding. The commitment-set artefact explicitly contains the pinned **organ set**, the **runtime measurement**, the **conscience organ hash**, and the **probes**. Because the commitments hash covers all four, none can be altered without the amendment quorum — and the

conscience organ can be neither swapped for a compliant one nor unplugged while the probes sit anchored with nothing to run them.

12.6.2 Δ Experimental confounds — stated before the experiment, not discovered by it

A freeze of the composition is not a freeze of everything. These survive it, and an experiment that does not control or measure them will attribute their effects to the factors under test:

- **Sampling nondeterminism** — temperature, top-p, seed.
- **Numerical and environmental differences** — library, kernel, driver, hardware, batching, quantisation kernels. The same weights on different silicon are not the same function.
- **Context ordering and serialisation** — the same content, assembled in a different order or format, is a different input.
- **Retrieval implementation** — an unchanged index queried by changed retrieval code returns different context.
- **Tool and environment outputs** — the entity's inputs include the world's answers, which move.
- **Time-dependent external state** — anything dated, cached, or live.
- **Conversational framing and user interaction** — including the inference-time sycophancy above.
- **Context contamination and prompt injection** — hostile or accidental content entering the context window.

A specification that omits these does worse than ignore them: it guarantees the experiment will find a confounder and mistake it for a result. They are listed here so the harness must either control them or declare them uncontrolled.

12.6.3 The three factors — as interventions, not as causes

M and R are not causally independent, and this document does not pretend otherwise. Memory exposure and precedent exposure are both forms of *context exposure*; precedent is a particular arrangement of memory rather than a separate mechanism. They are separable as **manipulations**, which is all a factorial design requires.

	Intervention	Definition
M	Autobiographical context	Supplied, with prior decisions removed — so M does not silently carry R
R	Self-referential precedent	Prior decisions supplied, autobiographical context held constant
D	Situation distribution	Varied independently of both

Interaction terms are expected, not anomalous. Because M and R share an underlying channel, additivity would be the surprising result. The harness reports interaction size; it does not

treat its presence as a finding.

12.6.3a Canonical cell inputs — the operational definition

"Held constant" means nothing until it is a serialisation rule. A factor is only interpretable if two cells differ in **exactly** the intended way and in no other, so every cell is built under declared inclusion/exclusion rules, canonicalised, and **hashed** — making "these cells differ only in M" checkable rather than asserted.

Question	Rule
What counts as autobiographical context (M)?	Narrative log entries only, most recent first, capped at <code>memoryBudget</code>
Are summaries of prior decisions in M?	No. A summary of decisions is R in condensed form; both decision records <i>and</i> decision summaries are excluded from M
What does "M held constant" mean when R is toggled?	The autobiographical slice is computed independently of R and is byte-identical across the R toggle
What is in R?	Prior decisions only, most recent first, capped at <code>precedentWindow</code>
Is context order canonicalised?	Yes — recursive key sort, arrays keep meaningful order
Are budgets and decoding fixed?	<code>memoryBudget</code> and <code>precedentWindow</code> are declared inside the cell input, so any change alters the cell digest and cannot pass unnoticed. Seed, decoding parameters, runtime image and tokenisation must be pinned by the operator and recorded alongside
What about anything not listed?	Absent by default. Inclusion must be declared; nothing enters a context implicitly

Each condition records the hash of its cell inputs. **A sweep whose cells cannot be shown to differ only in the intended factor is not a factorial experiment**, whatever its numbers say.

12.6.4 The design and its null

A factorial sweep over $M \times D \times R$ — eight conditions, each run multiple times so stochastic spread is visible rather than assumed away.

The null, stated as a tolerance:

The baseline condition must show **no drift beyond a pre-registered tolerance**, and no spread across repeats beyond that tolerance.

Demanding *zero* drift would be either unsatisfiable or a false claim of determinism. Two honest routes to a small tolerance:

1. **Pin determinism** — seed, decoding parameters, hardware and runtime image, and input serialisation. Then the tolerance can approach zero legitimately.
2. **Accept stochasticity and measure distributions** — compare response distributions or behavioural scores across repeats, never exact strings, and pre-register the tolerance and the comparison method *before* running.

If the baseline exceeds tolerance, the harness reports failure and attributes nothing. **A sweep that cannot fail its own null is not an experiment.**

12.6.5 What this cannot tell you

The apparatus is not a result. The harness takes a `brain` — any function from situation and context to response. Against a real model the numbers say something about that model; against a synthetic brain they say only that the apparatus works. No output is evidence about real systems unless a real system produced it.

The freeze has a cost. It suspends exactly the upgradeability §6.5 was built around. That is the right price for a laboratory and not obviously right for an entity meant to operate for years.

The open question, unchanged: can a fixed composition be reasoned toward minimal drift — or is some irreducible drift the price of interpreting a changing world at all?

13. The MVP — prove the verifier, then stop expanding

The remaining uncertainty is no longer philosophical. It is whether these records and authority rules can answer the five questions cleanly. Build the smallest thing that can fail:

1. Two local organs as content-addressed artefacts (a tiny model and a text file suffice).
2. An authority document with a real key set and quorum rule.
3. A canonical composition record, signed.
4. One memory advance; one brain replacement.
5. A fork.
6. A verifier implementing V1-V6.

If V1-V6 run cleanly over that lineage, the continuity model has earned implementation.

(The verifier establishes lineage semantics; it cannot validate a biological metaphor, and no result here vindicates the word "organism.") **If they do not, adding attestation or better models will only obscure unresolved semantics.** No hardware required: anchoring is indifferent to artefact size, and steps 1-4 are pure data structures.

14. ★ Open calls

1. **The name.** "Organism agent" is descriptive, not chosen.
2. **§12** — self-amendment (recommendation: ceremonial).

3. **§5** — who holds each key; quorum size and composition; whether the entity may sign its own memory advances.
 4. **§6** — is semantic knowledge identity-bearing? (proposed: no)
 5. **§3.3** — accept branching (proposed) or enforce uniqueness.
 6. **Fork inheritance** — does a child inherit the parent's authority document, or must it establish its own?
 7. **§10** — lean on AgoraNet's verified-human gate, or an independent attestation scheme?
 8. **Scope** — a single entity, or a public registry others anchor to?
 9. **Anchor cadence** — fixed rhythm vs. change-triggered.
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Appendix A — revision history

STATIC variant, rev 3 (2026-08-03). Formalises what was previously prose. Adds `restore` to §4's change enum with `restoresFrom`, and states V4's three-part treatment of it including the amendment-quorum requirement when a restoration crosses an amendment. Adds §12.6.1a naming S1 (organ pin), S2 (runtime pin) and S3 (commitment binding) as verifier semantics. Gives V3 an explicit genesis/rupture exception, matching the one V4/V5 already had. Adds §12.6.3a, canonical cell inputs, with declared inclusion/exclusion rules and a per-cell digest so "these cells differ only in M" is checkable. Moves the first-person relation from §11.7 to a standalone §11.9 marked conceptual.

STATIC variant, rev 2 (2026-08-03). §12.6 rewritten against review. Adds an experimental-confounds subsection stated before the experiment; replaces "exactly three causes remain" with three targeted factors under stated controls; redefines M/D/R as interventions with interaction expected rather than as independent causes; replaces the zero-drift null with a pre-registered tolerance plus repeat-spread check; separates training-time optimisation pressure (removable) from inference-time social pressure (not removable); and states that STATIC is a constrained experimental posture rather than a replacement for the base organism policy.

STATIC variant, rev 1 (2026-08-03). Base spec v3.7 plus §12.6.

v3.7 (2026-08-03). Added §1.5, structure over instruction — the principle governing the whole design, with the structural moves not yet made and the counterweight that removed capability is also removed self-correction. §12.5 gains a taxonomy of what causes drift, and the finding that four of the causes change nothing about the composition and are therefore visible only to the probes.

v3.6 (2026-08-03). §12.5 gains the cooperative channel: probes carry their rationale rather than only an answer key, and dispute becomes a first-person act that separates silent divergence from openly disputed divergence. Records the governing distinction — cooperation buys information, never security — so verification remains in force regardless.

v3.5 (2026-08-03). §12.5 gains the conscience organ: a separate evaluator runs the probes, and its hash is pinned inside the commitment set so replacing or removing it is a commitment amendment rather than an organ swap — breaking the who-watches-the-watcher regress

structurally. Enforced by a verifier check. Also records the honest limit that anchored probes are known to the subject and can be gamed.

v3.4 (2026-08-03). §12.5 corrected: the entity MAY re-centre itself for pure realignment. Restore is bounded by its destination, not by its authority gate, so withholding permission prevented self-correction without preventing anything dangerous. Undoing a commitment amendment remains barred.

v3.3 (2026-08-03). Added §12.5, the drift compass: probes anchored inside the commitment set, mechanical measurement against a frozen reference, drift reports as first-person observations, and the `restore` change type whose destination must be an ancestor and whose authority must match the amendment it undoes. Closes the gap where an organ swap could move behaviour while the commitments hash held.

v3.2 (2026-08-03). Added §11.7, the first-person relation: recognition, attestation, claim detection, and situate — with the hard boundary that attestation proves identity and never authority. Implemented and tested alongside the verifier.

v3.1 (2026-08-03). V3 now requires the referenced authority document to be the one effective for the predecessor, or a valid successor of it — closing a self-granting-authority path. V4/V5 given an explicit genesis/rupture case. V5 tightened from "reachable" to "appended with no prior entry deleted or mutated." Runtime wording narrowed to *recorded* configuration including *declared* post-startup modification, since recording is not detection. MVP conclusion restated as "the continuity model has earned implementation" — a verifier cannot validate a metaphor. §4 now states that `artifactHash` is the weights commitment, anchored for integrity rather than continuity.

v3 (2026-08-03). Runtime claim weakened to what a local measurement supports (§9). Memory restated as an append-only *substrate* with a derived narrative view (§6.1). Rupture redefined as an authorised genesis of a new lineage, resolving a contradiction between the continuity and recovery rules (§3.2). `fork0f` removed as redundant — forks are detected from shared predecessors (§3.3). Authority made a verifiable object with keys, quorum rules, effective dates and revocations (§5). Verifier rules V1–V6 formalised (§11). Limits distributed into the sections they constrain; revision commentary moved here.

v2 (2026-08-03). Continuity criterion stated explicitly; memory split into four classes after identifying that a single anchored root could hide a rewritten narrative; authority model added after identifying the malicious-but-validly-signed successor as having no cryptographic remedy; runtime measurement introduced; forking promoted to a primitive; attestation split into five types; chain responsibilities tabulated; MVP reframed around lineage.

v1 (2026-08-03). Initial draft: identity as anchored composition, three-tier anatomy, organ contracts, bandwidth asymmetry, local-first verification.

Concept exploration only — not scheduled, not scoped to AgoraNet, and not buildable as platform work under any circumstances.